Lesson 12: Introduction to Hypothesis Testing

BSTA 551: Statistical Inference

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Review: Where We Left Off

Lessons 9-11: Bootstrap Methods and Confidence Intervals

- Bootstrap principle: resample from sample to estimate sampling distribution
- Bootstrap percentile and t intervals
- Relationship between confidence intervals and plausible parameter values

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The Next Step: What if we want to make a **decision** rather than just estimate?

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Today's Goals:

1. Understand the hypothesis testing framework
2. Define null and alternative hypotheses
3. Learn about test statistics and rejection regions
4. Understand Type I and Type II errors
5. Apply these concepts to medical research examples

From Estimation to Testing

Confidence Intervals

- "What is a plausible range for the parameter?"
- Example: "95% CI for mean reduction in blood pressure is (8, 16) mmHg"
- Provides a range of values

Hypothesis Testing

- "Is there evidence that the parameter differs from a specific value?"
- Example: "Does this drug reduce blood pressure at all?"
- Provides a yes/no decision

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Key Insight (Devore 9.1)

Hypothesis testing is about deciding which of two contradictory claims about a parameter is correct based on sample data.

The Hypothesis Testing Framework

What is a Statistical Hypothesis?

Definition (Devore 9.1)

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Examples:

- The claim $\mu = 120$ mmHg, where μ is the true average systolic blood pressure
- The statement $p < 0.10$, where p is the proportion of patients who experience adverse events
- The assertion $\mu_{\text{treatment}} - \mu_{\text{control}} > 0$

The Two Competing Hypotheses

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Definitions (Devore 9.1)

Null Hypothesis (H_0): The claim that is initially assumed to be true (the "prior belief" or "status quo" claim).

Alternative Hypothesis (H_a or H_A): The assertion that is contradictory to H_0 — what we might conclude if the data provides strong evidence against H_0.

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> **Alternative Hypothesis (H_a or H_A):** The assertion that is contradictory to H_0 — what we might conclude if the data provides strong evidence against H_0.

The null hypothesis will be rejected in favor of the alternative **only if** sample evidence strongly suggests that H_0 is false.

The Courtroom Analogy

Criminal Trial (US)

- **Initial assumption:** Defendant is innocent
- **Burden of proof:** On prosecution
- **Standard:** “Beyond reasonable doubt”
- **Decision:** Guilty or Not Guilty

Note: “Not Guilty” $\neq$ “Innocent”

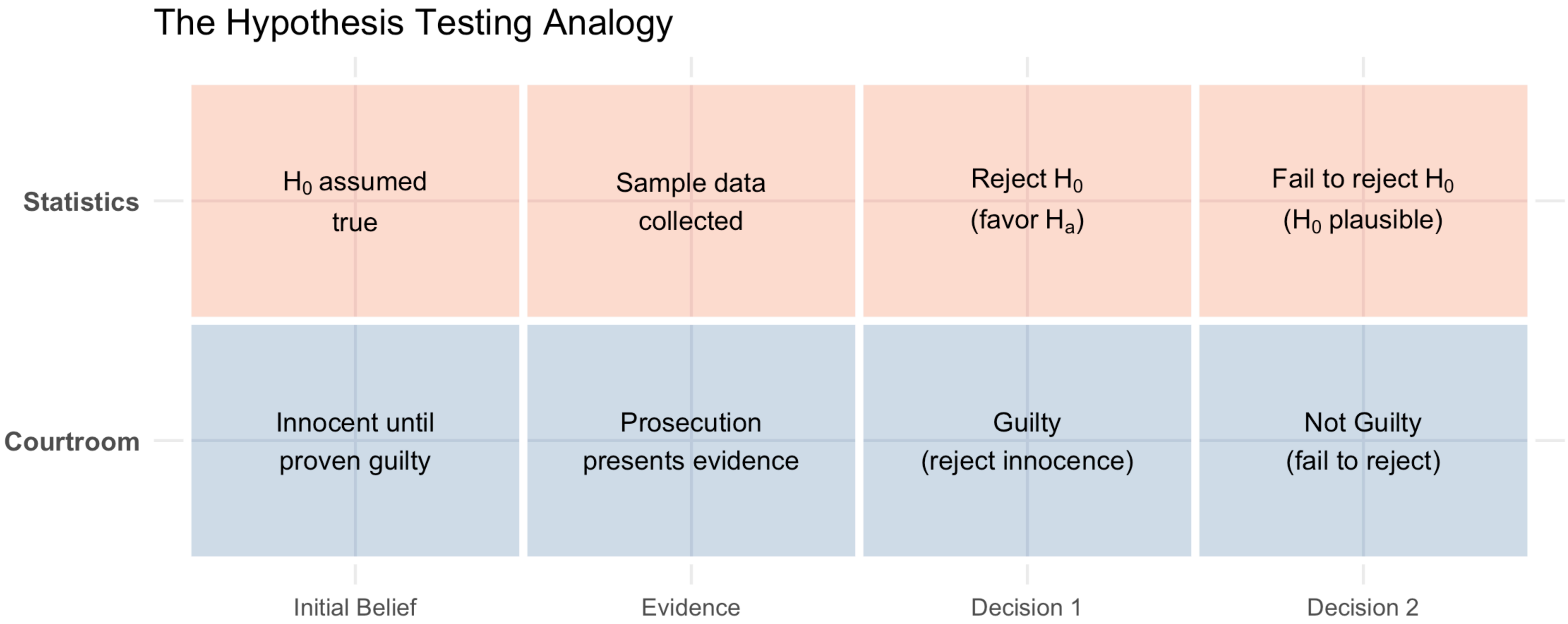
Hypothesis Testing

- **Initial assumption:** H_0 is true
- **Burden of proof:** On data to contradict H_0
- **Standard:** Significance level α
- **Decision:** Reject H_0 or Fail to Reject H_0

Note: “Fail to Reject” $\neq$ “H_0 is True”

Visualizing the Analogy

▶ Code



Two Possible Conclusions

Important Distinction

A hypothesis test can only lead to one of two conclusions:

1. **Reject** H_0 — Evidence strongly suggests H_0 is false
2. **Fail to reject** H_0 — Evidence does not strongly contradict H_0

We **never** say "Accept H_0" because failing to reject is not the same as proving H_0 is true!

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Analogy: A "Not Guilty" verdict doesn't prove innocence — it means there wasn't enough evidence to convict.

Formulating Hypotheses

Example 1: New Antihypertensive Drug

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Example 1: New Antihypertensive Drug

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Hypotheses:

$$H_0 : \mu = 10 \quad \text{(no improvement over standard)}$$

$$H_a : \mu > 10 \quad \text{(new drug is better)}$$

The company will only switch to the new drug if there is **convincing evidence** of improvement.

Example 2: Is Normal Body Temperature Really 98.6°F?

The widely accepted "normal" human body temperature of 98.6°F (37°C) dates back to a 19th century German study. Recent research suggests this value may be inaccurate.

Let μ denote the true mean body temperature of healthy adults.

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Hypotheses:

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$$H_a : \mu \neq 98.6 \quad \text{(traditional value is wrong — either direction)}$$

This is a **two-sided** test because departures in either direction would be scientifically important.

Example 3: Adverse Event Rate

An investigational drug is considered safe if the rate of serious adverse events is at most 5%. In a clinical trial, researchers want to determine if the drug exceeds this threshold.

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$$H_0 : p = 0.05 \quad \text{(drug meets safety threshold)}$$

$$H_a : p > 0.05 \quad \text{(drug exceeds threshold — unsafe)}$$

This is a **one-sided** (upper-tailed) test because only a rate higher than 5% would be concerning.

General Structure of Hypotheses

The null hypothesis is typically stated as an **equality**:

$$H_0 : \theta = \theta_0$$

where θ_0 is called the **null value** of the parameter.

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The alternative hypothesis takes one of three forms:

Alternative	**Test Type**	**Example Context**
$H_a : \theta > \theta_0$	Upper-tailed	New treatment better than standard
$H_a : \theta < \theta_0$	Lower-tailed	Reduced error rate
$H_a : \theta \neq \theta_0$	Two-tailed	Testing for any difference

Choosing the Alternative: Research vs. Null

Practical Guidance (Devore 9.1)

- H_a is often called the **research hypothesis** — what the investigator wants to demonstrate
- H_0 represents the **status quo** or “no effect/no difference”
- The burden of proof is on showing H_a is true

This is **conservative**: we only claim an effect when the data provides strong evidence.

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Medical Research Context:

- New treatment must **prove** it's better than standard
- We protect against false claims of efficacy
- FDA approval requires strong evidence that benefits outweigh risks

Test Procedures

Components of a Test Procedure

Definition (Devore 9.1)

A **test procedure** is specified by:

1. A **test statistic**: a function of the sample data used to make the decision
2. A **rejection region** (critical region): the set of test statistic values for which H_0 will be rejected

The null hypothesis is rejected if and only if the observed test statistic value falls in the rejection region.

Example: Testing Vaccine Efficacy

A new vaccine is tested. Let p be the true proportion of vaccinated individuals who develop the disease. We test $H_0 : p = 0.20$ vs $H_a : p < 0.20$.

In a trial of $n = 200$ vaccinated individuals:

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In a trial of $n = 200$ vaccinated individuals:

Test Statistic: X = number who develop disease

Rejection Region: Reject H_0 if $X \leq 30$

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Test Statistic: X = number who develop disease

Rejection Region: Reject H_0 if $X \leq 30$

```r
# Under H0: X ~ Binomial(200, 0.20)
# Expected value under H0
n <- 200
p0 <- 0.20
expected_X <- n * p0
cat("Expected X under H0:", expected_X, "\n")
```

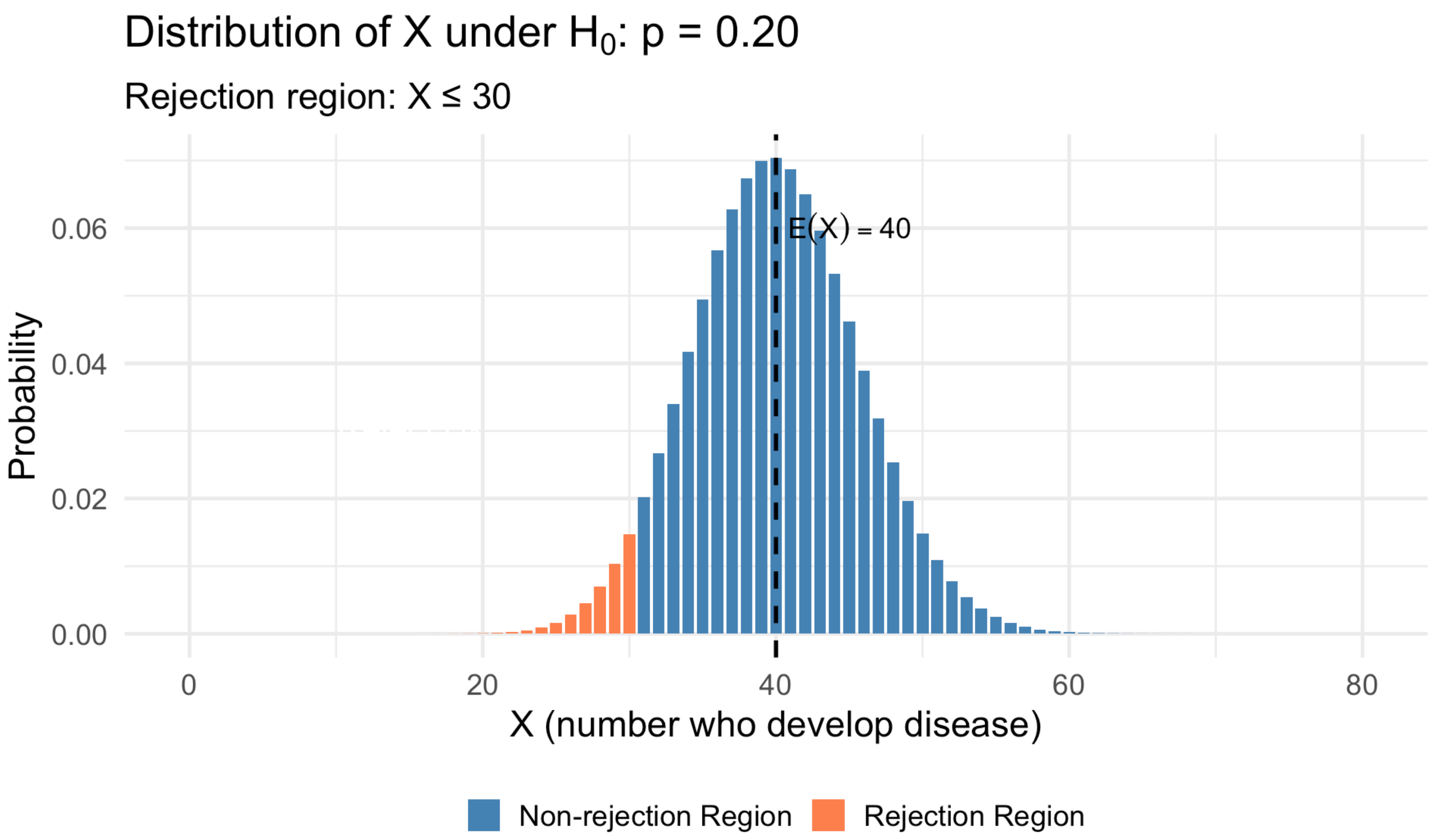
```
Expected X under H0: 40
```

```r
cat("Rejection region: X ≤ 30 (substantially less than 40)")
```

```
Rejection region: X ≤ 30 (substantially less than 40)
```

Visualizing the Rejection Region

▶ Code



Example: Testing Engine Fuel Efficiency (Devore Example 9.3)

A manufacturer considers modifying truck engines to improve fuel efficiency. Current mean: 25 mpg, SD: 3 mpg (known). Let μ = true mean mpg for modified engines.

$$H_0 : \mu = 25 \quad \text{vs} \quad H_a : \mu > 25$$

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Test with $n = 10$ trucks:

Test Statistic: $\bar{X}$ = sample mean fuel efficiency

Possible Rejection Region: Reject H_0 if $\bar{X} \geq 27.5$

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Test with $n = 10$ trucks:

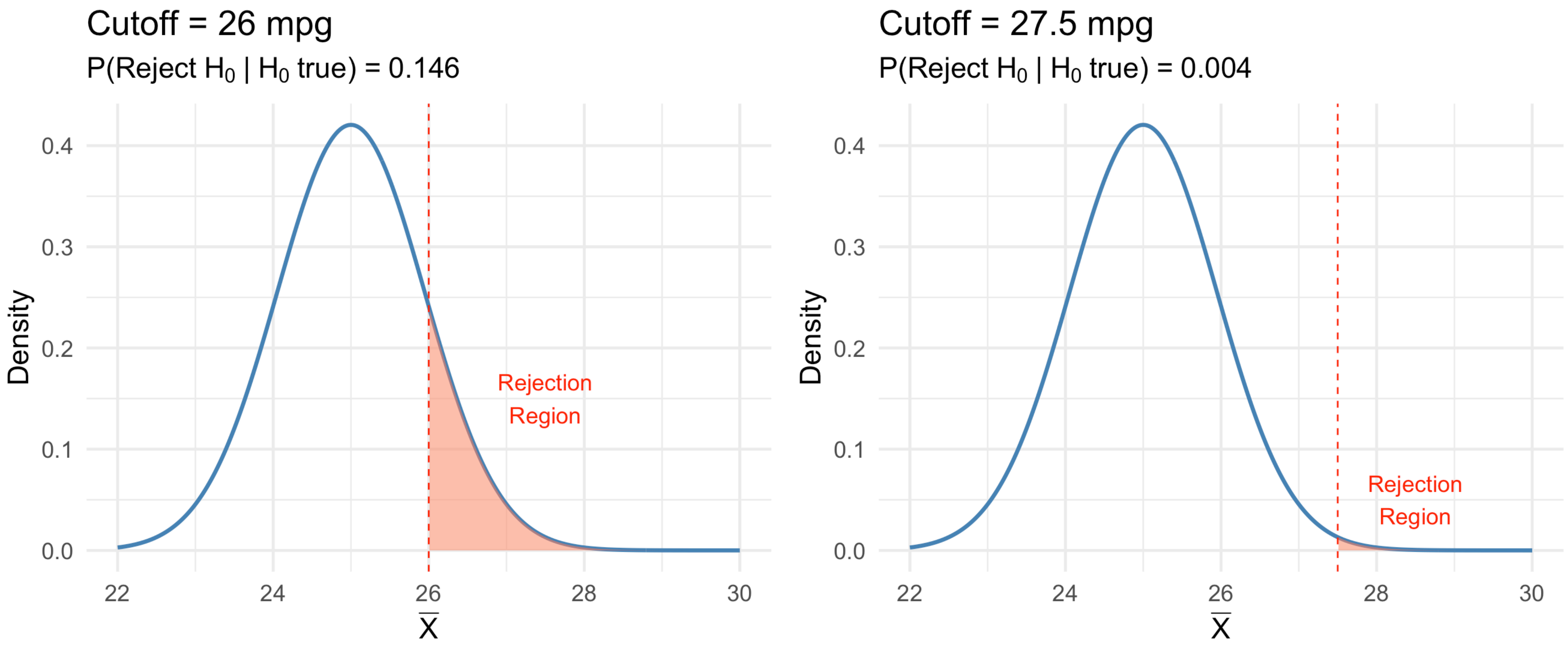
Test Statistic: $\bar{X}$ = sample mean fuel efficiency

Possible Rejection Region: Reject H_0 if $\bar{X} \geq 27.5$

Key Question: How do we choose the cutoff value (27.5 vs 26 vs 28)?

The Tradeoff in Choosing the Cutoff

▶ Code



Lower cutoff: More likely to reject H_0 (even when true!)

Higher cutoff: Less likely to reject H_0 (even when false!)

Type I and Type II Errors

Two Types of Errors

When making a decision based on sample data, two types of errors are possible:

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Definitions (Devore 9.1)

Type I Error: Rejecting H_0 when it is actually **true** (false positive)

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	H_0 **True**	H_0 **False**
Reject H_0	**Type I Error**	Correct Decision
Fail to Reject H_0	Correct Decision	**Type II Error**

Error Probabilities

Notation

$$\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

$$\beta = P(\text{Type II Error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

Power = $1 - \beta$ = $P(\text{Reject } H_0 \mid H_0 \text{ is false})$

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- α is called the **significance level** of the test
- Common choices: $\alpha = 0.05, \alpha = 0.01, \alpha = 0.10$

Medical Context: Errors Have Consequences

Example: Testing whether a new cancer drug is effective

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Type I Error (False Positive):

- Conclude drug works when it doesn't
- Patients receive ineffective treatment
- Resources wasted on worthless drug
- *Consequence:* Harm to patients, wasted resources

Medical Context: Errors Have Consequences

Example: Testing whether a new cancer drug is effective

Type I Error (False Positive):

- Conclude drug works when it doesn't
- Patients receive ineffective treatment
- Resources wasted on worthless drug
- *Consequence:* Harm to patients, wasted resources

Type II Error (False Negative):

- Conclude drug doesn't work when it does
- Effective treatment never reaches patients
- *Consequence:* Missed opportunity to save lives

Example: Drug Approval Errors (Devore Example 9.4)

Scenario: Testing a new antihypertensive drug against standard care.

$H_0 : \mu_{\text{new}} = \mu_{\text{standard}}$ (no improvement)

$H_a : \mu_{\text{new}} > \mu_{\text{standard}}$ (new drug is better)

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Error Type	What Happens	Consequence
Type I	Approve ineffective drug	Patients get worse care; company invested in useless drug
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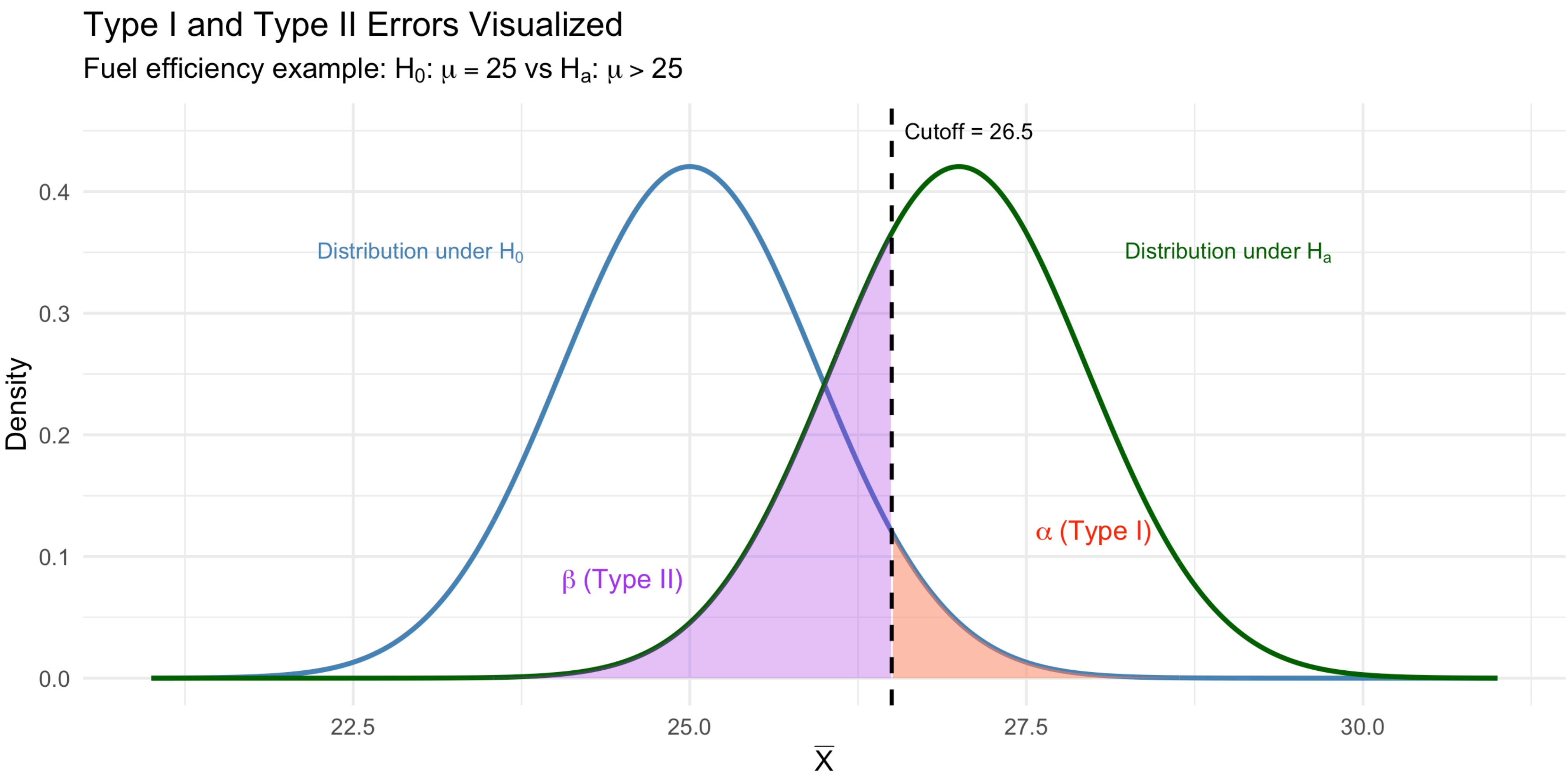
Error Type	What Happens	Consequence
Type I	Approve ineffective drug	Patients get worse care; company invested in useless drug
Type II	Reject effective drug	Patients miss beneficial treatment; opportunity loss

FDA Approach: Keep α small (typically 0.05) — protect against approving ineffective drugs.

Visualizing Type I and Type II Errors

▶ Code



Computing Error Probabilities: Example

For the fuel efficiency test with cutoff $\bar{x} \geq 27.5$:

```r
mu0 <- 25
sigma <- 3
n <- 10
se <- sigma / sqrt(n)
cutoff <- 27.5

# Type I error: P(reject H0 | H0 true)
alpha <- 1 - pnorm(cutoff, mean = mu0, sd = se)
cat("α = P(X̄ ≥", cutoff, "| μ =", mu0, ") =", round(alpha, 4), "\n")
```

```
α = P(X̄ ≥ 27.5 | μ = 25 ) = 0.0042
```

```r
# Type II error: P(fail to reject | H0 false)
# Suppose true mean is μ = 26.5
mu_true <- 26.5
beta <- pnorm(cutoff, mean = mu_true, sd = se)
cat("β = P(X̄ <", cutoff, "| μ =", mu_true, ") =", round(beta, 4), "\n")
```

```
β = P(X̄ < 27.5 | μ = 26.5 ) = 0.8541
```

```r
cat("Power = 1 - β =", round(1 - beta, 4))
```

```
Power = 1 - β = 0.1459
```

The Fundamental Tradeoff

The Tradeoff (Devore 9.1)

For a **fixed sample size**, decreasing α (Type I error) necessarily **increases** β (Type II error), and vice versa.

The only way to decrease **both** error probabilities simultaneously is to **increase the sample size**.

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The only way to decrease **both** error probabilities simultaneously is to **increase the sample size**.

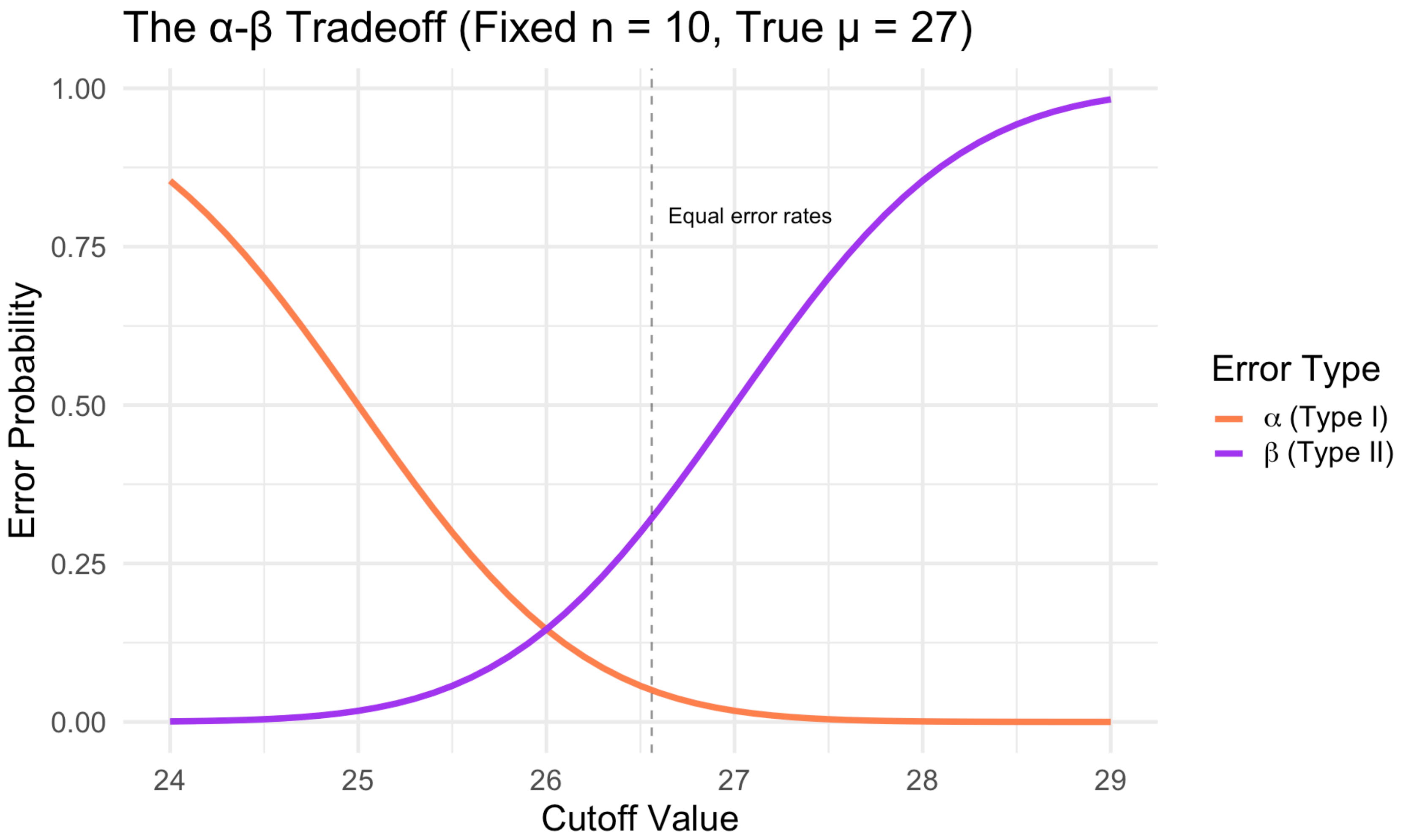
```r
# Demonstrate tradeoff with different cutoffs
cutoffs <- c(26, 26.5, 27, 27.5, 28)
alpha_vals <- 1 - pnorm(cutoffs, mu0, se)
beta_vals <- pnorm(cutoffs, mean = 27, sd = se)  # assuming true mu = 27

tibble(Cutoff = cutoffs, Alpha = round(alpha_vals, 4), Beta = round(beta_vals, 4)) |>
  mutate(Power = 1 - Beta)
```

```
# A tibble: 5 × 4
  Cutoff  Alpha  Beta Power
   <dbl>  <dbl> <dbl> <dbl>
1   26   0.146  0.146 0.854
2   26.5 0.0569 0.299 0.701
3   27   0.0175 0.5   0.5
4   27.5 0.0042 0.701 0.299
5   28   0.0008 0.854 0.146
```

Visualizing the Tradeoff

▶ Code



Simulation: Demonstrating Error Types

▶ Code

```
Simulated Type I Error Rate: 0.061
```

▶ Code

```
Theoretical α: 0.0569
```

▶ Code

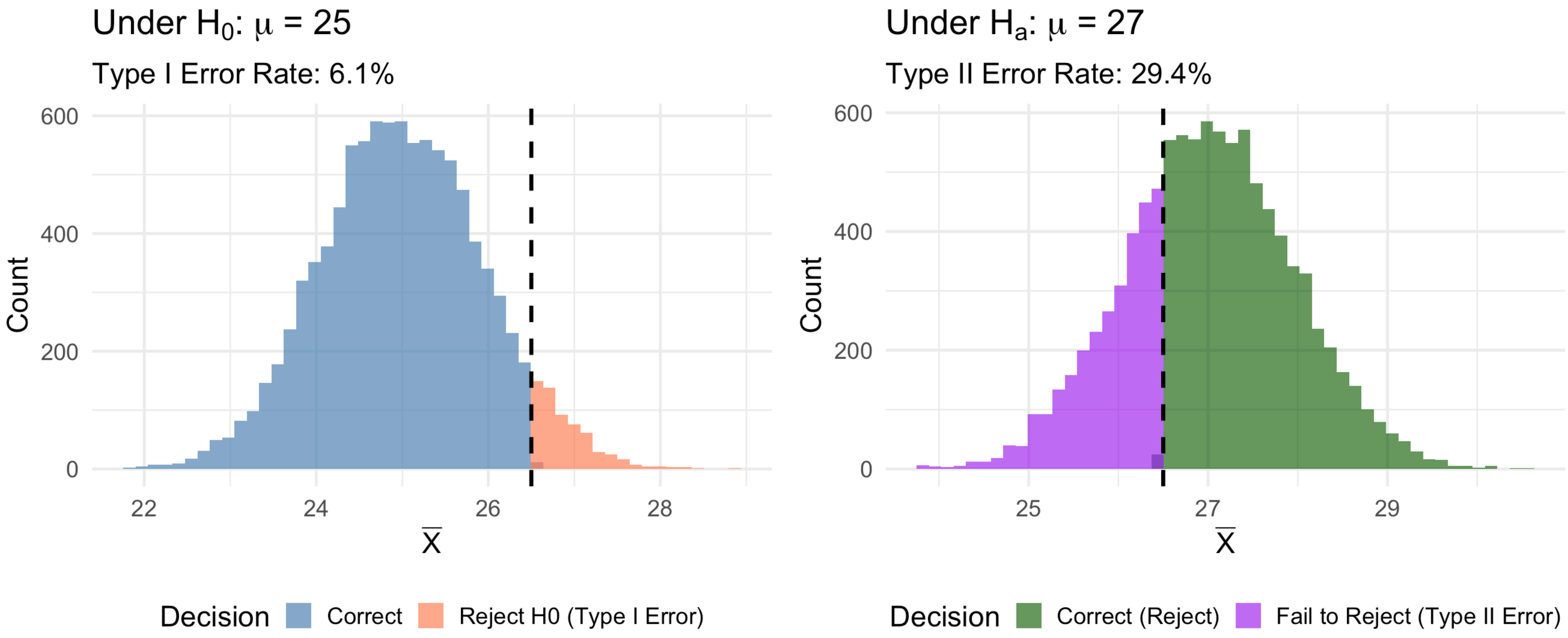
```
Simulated Type II Error Rate: 0.2939
```

▶ Code

```
Theoretical β: 0.2991
```

Visualizing Simulated Decisions

▶ Code



Significance Level and Rejection Regions

The Significance Level

Definition

The **significance level** α of a test is the probability of making a Type I error — rejecting H_0 when it is actually true.

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Common choices: $\alpha = 0.05, \alpha = 0.01, \alpha = 0.10$

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Definition

The **significance level** α of a test is the probability of making a Type I error — rejecting H_0 when it is actually true.

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Common choices: $\alpha = 0.05, \alpha = 0.01, \alpha = 0.10$

Why prioritize controlling Type I error?

- Type I errors can lead to **false claims** (e.g., ineffective drugs approved)
- Scientific community uses $\alpha = 0.05$ as a standard threshold
- α is chosen **before** collecting data

Constructing Rejection Regions

Once α is chosen, we determine the rejection region:

Upper-tailed test ($H_a : \mu > \mu_0$):

$$\text{Reject } H_0 \text{ if } \bar{X} \geq \mu_0 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$$

Lower-tailed test ($H_a : \mu < \mu_0$):

$$\text{Reject } H_0 \text{ if } \bar{X} \leq \mu_0 - z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$$

Two-tailed test ($H_a : \mu \neq \mu_0$):

$$\text{Reject } H_0 \text{ if } |\bar{X} - \mu_0| \geq z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Example: Calculating Critical Values

For the fuel efficiency test at $\alpha = 0.05$:

```r
alpha <- 0.05
mu0 <- 25
sigma <- 3
n <- 10
se <- sigma / sqrt(n)

# Critical value for upper-tailed test
z_alpha <- qnorm(1 - alpha)
critical_value <- mu0 + z_alpha * se

cat("z_α =", round(z_alpha, 3), "\n")
```

```
z_α = 1.645
```

```r
cat("Critical value: X̄ ≥", round(critical_value, 3), "\n")
```

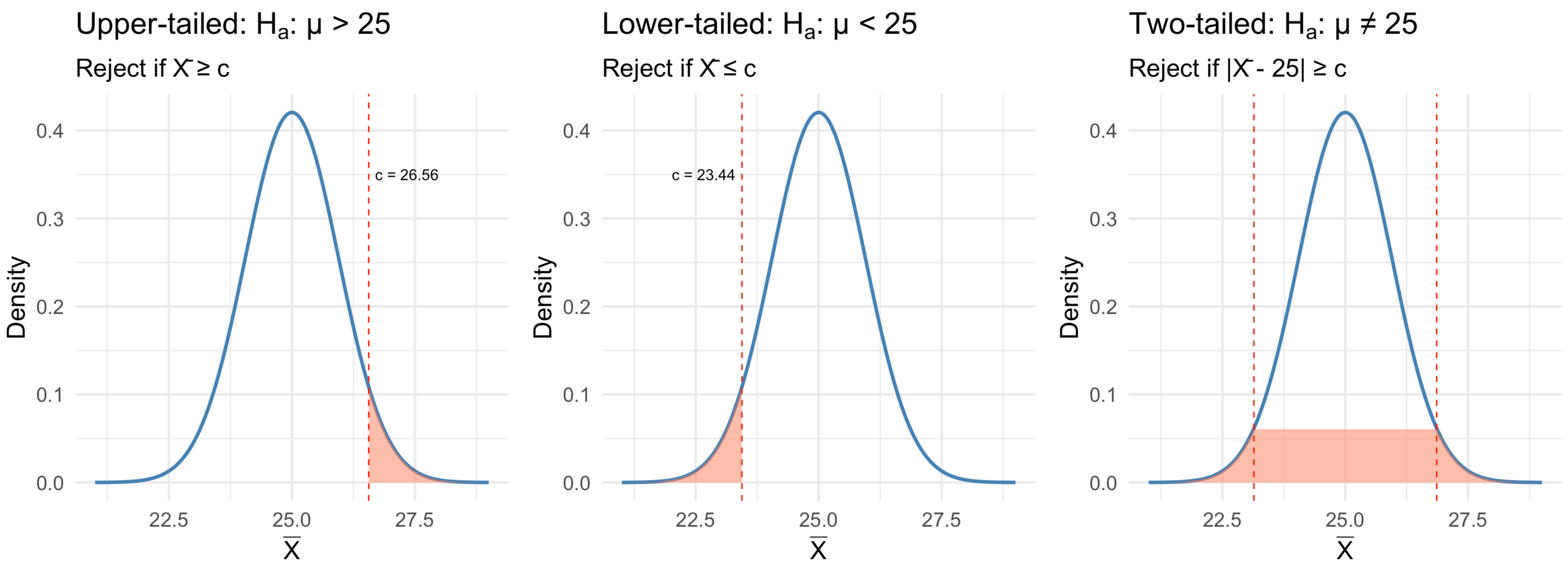
```
Critical value: X̄ ≥ 26.56
```

```r
cat("Rejection region: X̄ ≥", round(critical_value, 3))
```

```
Rejection region: X̄ ≥ 26.56
```

Visualizing Critical Regions by Tail Type

▶ Code



Medical Research Example: Clinical Trial

Complete Example: Blood Pressure Drug Trial

Setting: Testing a new antihypertensive drug.

Data: $n = 25$ patients, mean BP reduction $\bar{x} = 12.5$ mmHg, known population SD $\sigma = 8$ mmHg.

Question: Is there evidence that the true mean reduction exceeds 10 mmHg?

Complete Example: Blood Pressure Drug Trial

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Data: $n = 25$ patients, mean BP reduction $\bar{x} = 12.5$ mmHg, known population SD $\sigma = 8$ mmHg.

Question: Is there evidence that the true mean reduction exceeds 10 mmHg?

Step 1: State Hypotheses

$$H_0 : \mu = 10 \quad \text{vs} \quad H_a : \mu > 10$$

Step 2: Choose significance level

$$\alpha = 0.05$$

Step 3: Identify Test Statistic and Rejection Region

```r
# Given data
xbar <- 12.5
sigma <- 8
n <- 25
mu0 <- 10
alpha <- 0.05

# Standard error
se <- sigma / sqrt(n)

# Test statistic: Z = (X̄ - μ₀) / (σ/√n)
z_stat <- (xbar - mu0) / se
cat("Test statistic: z =", round(z_stat, 3), "\n")
```

```
Test statistic: z = 1.562
```

```r
# Critical value for α = 0.05 (upper-tailed)
z_crit <- qnorm(1 - alpha)
cat("Critical value: z_α =", round(z_crit, 3), "\n")
```

```
Critical value: z_α = 1.645
```

```r
# Rejection region
cat("Rejection region: z ≥", round(z_crit, 3))
```

```
Rejection region: z ≥ 1.645
```

Step 4: Make Decision

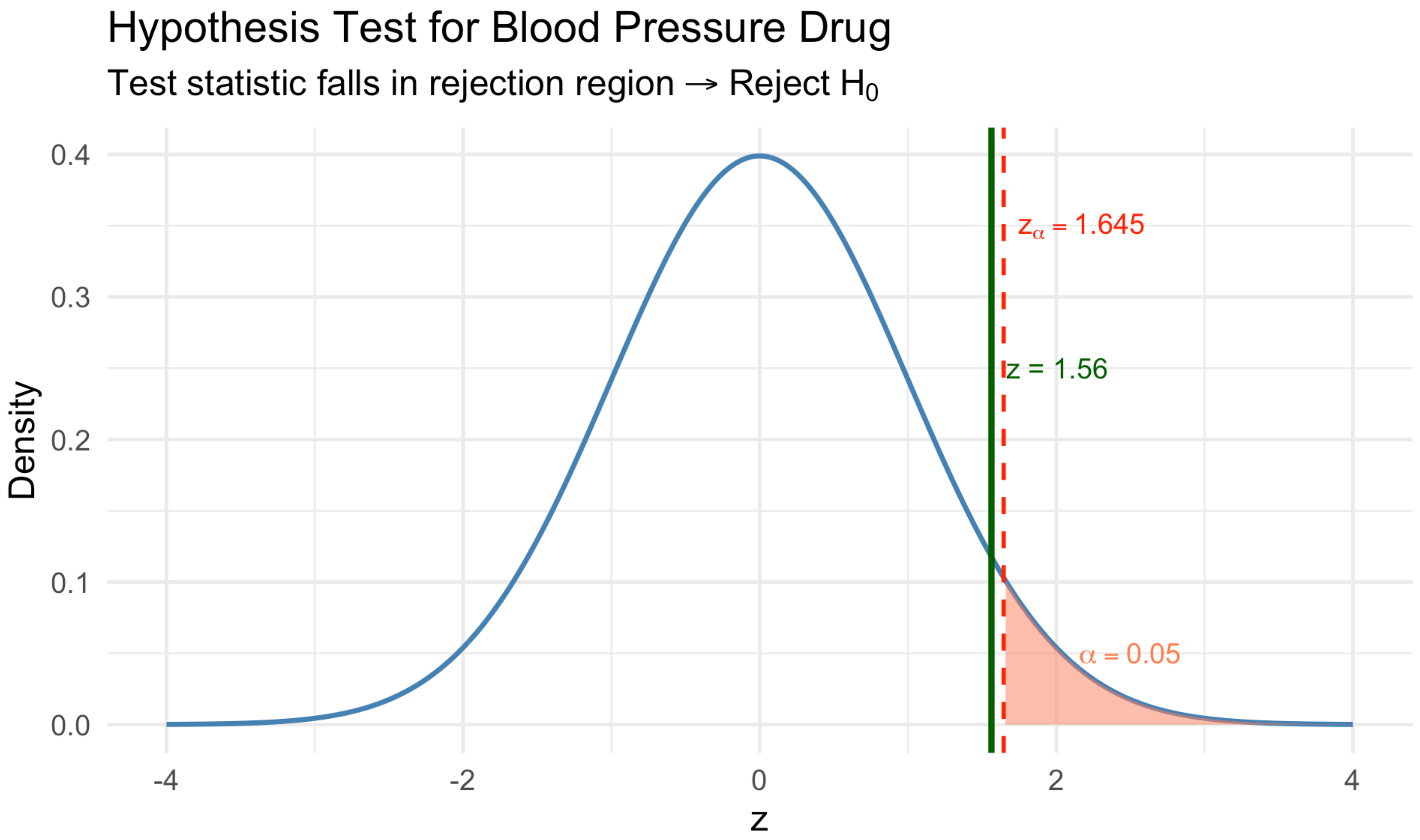
```
# Decision
if (z_stat >= z_crit) {
  cat("Decision: Reject H₀\n\n")
  cat("Interpretation: There is statistically significant evidence at α = 0.05\n")
  cat("that the true mean BP reduction exceeds 10 mmHg.")
} else {
  cat("Decision: Fail to reject H₀\n\n")
  cat("Interpretation: There is insufficient evidence at α = 0.05\n")
  cat("to conclude that the true mean BP reduction exceeds 10 mmHg.")
}
```

```
Decision: Fail to reject H₀

Interpretation: There is insufficient evidence at α = 0.05
to conclude that the true mean BP reduction exceeds 10 mmHg.
```

Visualizing the Test Result

▶ Code



Error Analysis for This Test

▶ Code

```
Type I Error Probability: α = 0.05
```

▶ Code

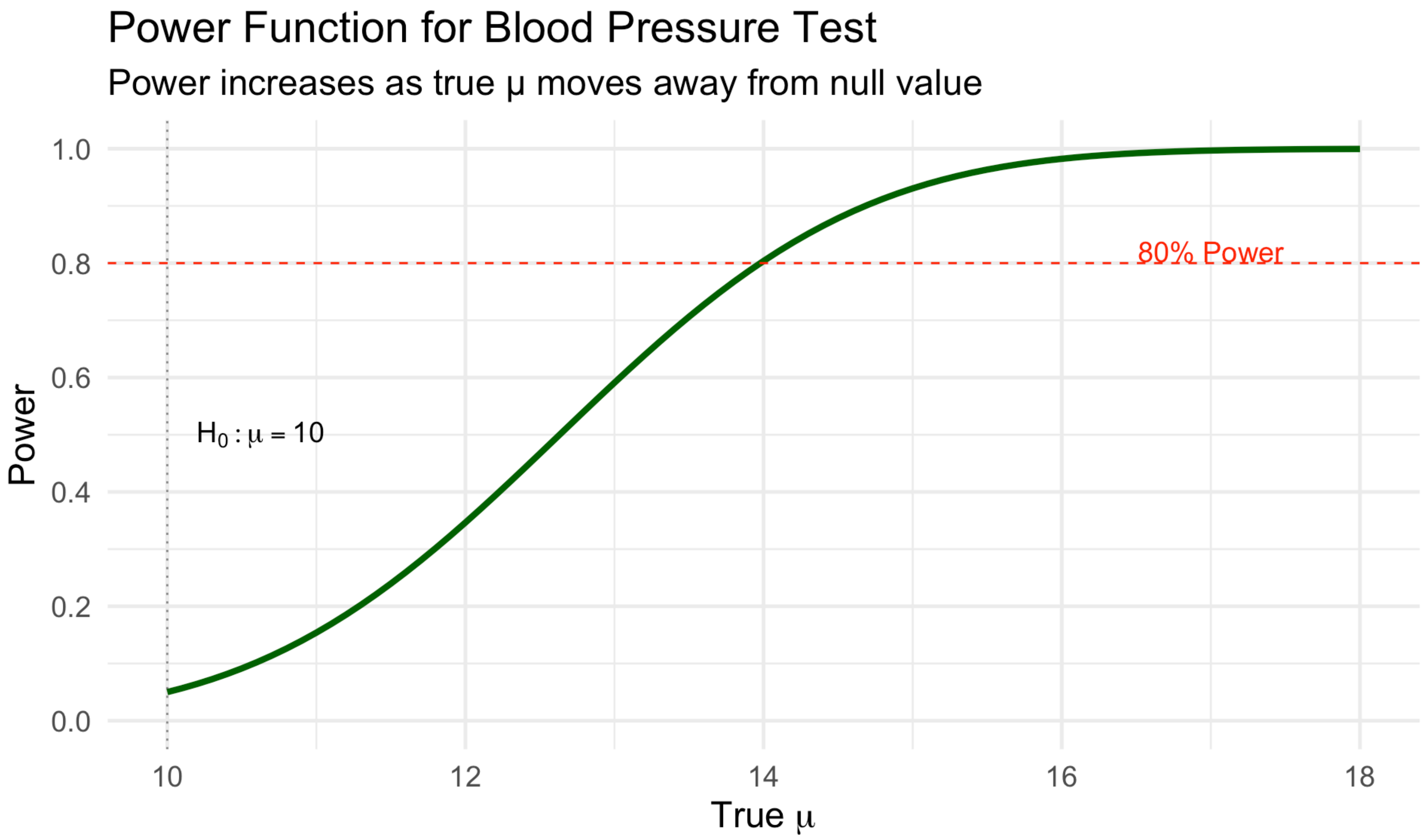
```
Type II Error and Power for different true μ values:
```

▶ Code

```
# A tibble: 5 × 3
  `True μ` `β (Type II)` `Power (1-β)`
     <dbl>         <dbl>         <dbl>
1       11        0.846          0.154
2       12        0.654          0.346
3       13        0.409          0.591
4       14        0.196          0.804
5       15        0.0694         0.931
```

Power Curve

▶ Code



Connection to Confidence Intervals

CIs and Tests: Two Sides of the Same Coin

Key Relationship

A $100(1-\alpha)\%$ confidence interval contains all parameter values that would **not** be rejected by a two-sided test at level α.

- If the CI **excludes** the null value θ_0 → Reject H_0
- If the CI **includes** the null value θ_0 → Fail to reject H_0

Example: Connecting CI and Test

For our blood pressure example:

```r
# 95% confidence interval (two-sided)
ci_lower <- xbar - qnorm(0.975) * se
ci_upper <- xbar + qnorm(0.975) * se

cat("95% CI for μ: (", round(ci_lower, 2), ",", round(ci_upper, 2), ")\n")
```

```
95% CI for μ: ( 9.36 , 15.64 )
```

```r
cat("Null value μ₀ = 10\n\n")
```

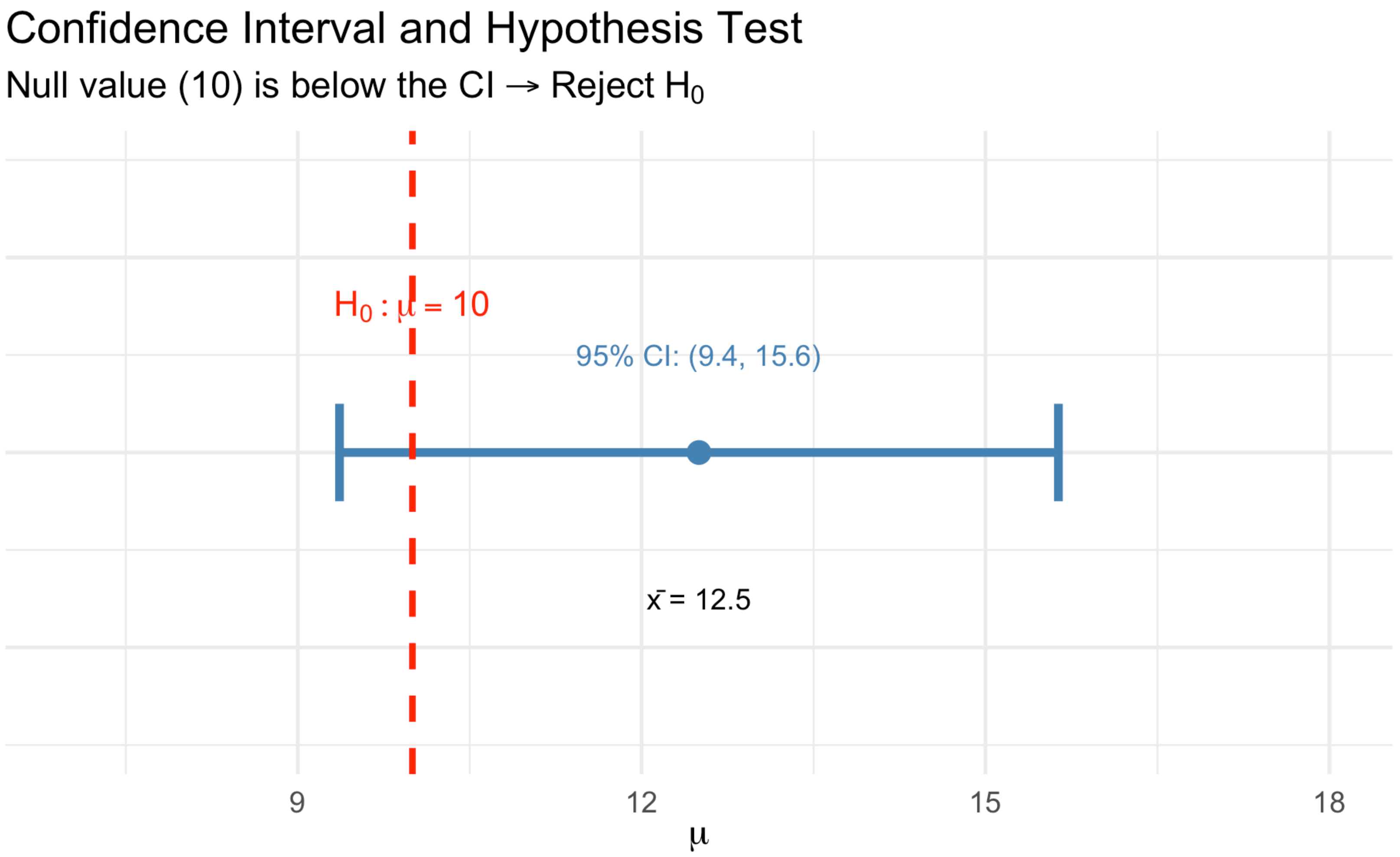
```
Null value μ₀ = 10
```

```r
if (ci_lower > 10) {
  cat("Since 10 is below the CI, we would reject H₀: μ = 10\n")
  cat("in favor of Hₐ: μ ≠ 10 at α = 0.05")
} else if (ci_upper < 10) {
  cat("Since 10 is above the CI, we would reject H₀: μ = 10")
} else {
  cat("Since 10 is within the CI, we would fail to reject H₀: μ = 10\n")
  cat("at α = 0.05")
}
```

```
Since 10 is within the CI, we would fail to reject H₀: μ = 10
at α = 0.05
```

CI-Test Duality Visualized

▶ Code



Summary and Key Takeaways

The Hypothesis Testing Framework

Core Components

1. **Null Hypothesis (H_0):** The status quo claim (initially assumed true)
2. **Alternative Hypothesis (H_a):** What we want to show (contradicts H_0)
3. **Test Statistic:** Function of sample data for making decisions
4. **Rejection Region:** Values of test statistic leading to rejection
5. **Significance Level (α):** Probability of Type I error

The Two Types of Errors

	H_0 **True**	H_0 **False**
Reject H_0	Type I Error (α)	Correct (Power)
Fail to Reject	Correct	Type II Error (β)

The Two Types of Errors

	H_0 **True**	H_0 **False**
Reject H_0	Type I Error (α)	Correct (Power)
Fail to Reject	Correct	Type II Error (β)

Key points:

- We control α by choosing it before the test
- β depends on the true parameter value
- Power = $1 - \beta$ = probability of correctly rejecting false H_0
- Tradeoff: Can't minimize both errors without increasing n

Connection to Previous Material

Topic	Lesson	Connection to Testing
Point Estimation	2-4	Test statistics are often MLEs
Sufficient Statistics	5	Best tests use sufficient statistics
Confidence Intervals	7-8	CIs give all non-rejected values
Bootstrap	9-11	Bootstrap hypothesis tests use resampling

Looking Ahead

Coming up in this course:

- **P-values:** Continuous measure of evidence against H_0
- **Power and sample size:** How many subjects do we need?
- **Tests for specific parameters:** Mean (z and t), proportion, variance
- **Two-sample tests:** Comparing populations
- **Bootstrap hypothesis tests:** Resampling-based testing

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapter 9.1
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 3.1-3.2, 8.1-8.2
- Wasserman, L. (2004). *All of Statistics*. Chapter 10.

Questions?

Thank you!